Markov Decision Process

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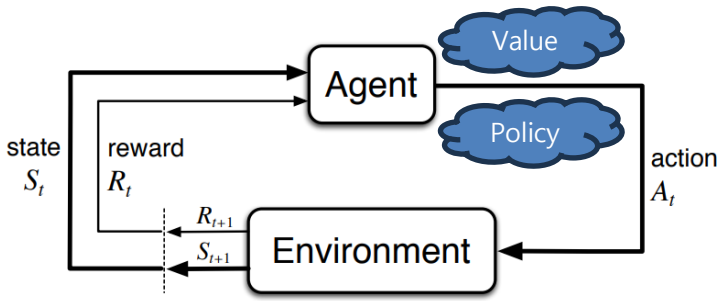
Agent-Environment Interface

- The agent-environment interaction in RL



Agent: learner, decision-maker (controller)

Environment: The thing which the agent interact with (plant or system)

State: $S_t \in \mathcal{S}$, $\mathcal{S}$ is the set of possible states

Action: $A_t \in \mathcal{A}(S_t)$, $\mathcal{A}(S_t)$ is the set of actions available in state S_t

Reward: $R_{t+1} \in \mathcal{R} \subset \mathbb{R}$, numerical values

Value: $v(S_t)$ or $q(A_t)$ measure of estimating how good the state or action is

Policy: $\pi_t = \pi_t(a|s)$, probability that $A_t = a$ if $S_t = s$
mapping from states to probabilities of each possible action

⇒ RL determines agent's policy to maximize the total amount of rewards

Agent-Environment Interface

(Ex) Recycling Robot

A mobile robot has the job of collecting empty soda cans in an office environment. It has sensors for detecting cans, and an arm and gripper that can pick them up and place them in an onboard bin; it runs on a rechargeable battery. The robot's control system has components for interpreting sensory information, for navigating, and for controlling the arm and gripper. High-level decisions about how to search for cans are made by a reinforcement learning agent based on the current charge level of the battery. This agent has to decide whether the robot should (1) actively search for a can for a certain period of time, (2) remain stationary and wait for someone to bring it a can, or (3) head back to its home base to recharge its battery. This decision has to be made either periodically or whenever certain events occur, such as finding an empty can. The agent therefore has three actions, and its state is determined by the state of the battery. The rewards might be zero most of the time, but then become positive when the robot secures an empty can, or large and negative if the battery runs all the way down. In this example, the reinforcement learning agent is not the entire robot. The states it monitors describe conditions within the robot itself, not conditions of the robot's external environment. The agent's environment therefore includes the rest of the robot, which might contain other complex decision-making systems, as well as the robot's external environment.

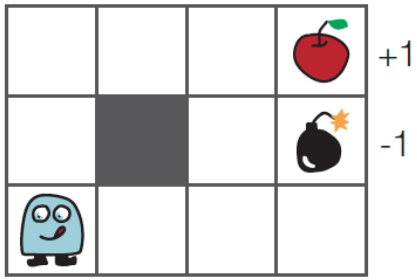
$$\mathcal{S} = \{\texttt{high}, \texttt{low}\}$$
$$\mathcal{A}(\texttt{high}) = \{\texttt{search}, \texttt{wait}\}$$
$$\mathcal{A}(\texttt{low}) = \{\texttt{search}, \texttt{wait}, \texttt{recharge}\}$$

$r_{\texttt{search}}$ = expected no. of cans while searching

$r_{\texttt{wait}}$ = expected no. of cans while waiting

Agent-Environment Interface

(Ex) 3x4 Grid World



$\mathcal{S} =$

$\mathcal{A} =$

$r =$

- 에이전트는 상하좌우 네 방향으로 이동할 수 있다.
- [그림 4-8]에서 회색 칸은 벽을 뜻하며 벽 안으로는 들어갈 수 없다.
- 그리드 바깥도 벽으로 둘러싸여 더 이상 나아갈 수 없다.
- 벽에 부딪히면 보상은 0이다.
- 사과는 보상 +1, 폭탄은 보상 −1, 그 외의 보상은 0이다.
- 환경의 상태 전이는 고유하다(결정적). 즉, 에이전트가 오른쪽으로 이동하는 행동을 선택하면 (벽만 없다면) 반드시 오른쪽으로 이동한다.
- 이번 문제는 일회성 과제로서, 사과를 얻으면 종료한다.

Tasks

- Episodic task (일회성 과제)
  - Terminal state 에서 종료되는 과제 ('끝' 이 있는 과제)
  - 게임, 바둑, Grid World 등

- Continuing task (지속적 과제)
  - Terminal state 가 없는 과제 ('끝' 이 없는 과제)
  - 재고관리, 로봇작업, Recycle Robot 등

Return

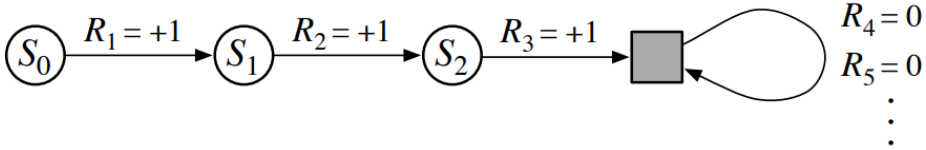
- Goal
  - Agent's goal is to maximize the cumulative reward it receives in the long run
- Return G_t (수익)
  - Sum of rewards after time step t
  - Episodic tasks

$$G_t = R_{t+1} + R_{t+2} + R_{t+3} + \cdots + R_T$$

  - Continuing tasks ($T = \infty$)

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots$$

    - γ : discount rate ($0 \le \gamma \le 1$)
    - For the continuing task, if $\gamma = 1$ then $G_t \to \infty$



Markov Property

- Definition
  - A state S_t is Markov if and only if

$$\Pr\{S_{t+1} \mid S_0, S_1, \cdots, S_t\} = \Pr\{S_{t+1} \mid S_t\}$$

    - The state captures all relevant information from the history
    - Once the state is known, the history may be thrown away
    - i.e. The state is a sufficient statistic of the future

    "The future is independent of the past given the present"

- In Reinforcement Learning

$$\Pr\{R_{t+1} = r, S_{t+1} = s' \mid S_0, A_0, R_1, \cdots, S_{t-1}, A_{t-1}, R_t, S_t, A_t\}$$

$$= \Pr\{R_{t+1} = r, S_{t+1} = s' \mid S_t, A_t\} \qquad \text{(Markov property)}$$

$$= p(s', r \mid s, a)$$

Markov Decision Processes

- Definition (MDP)
  - $(\mathcal{S}, \mathcal{A}, T, r, \gamma)$
  - A reinforcement learning task that satisfies the Markov property
  - Finite MDP if the state space $\mathcal{S}$ and action space $\mathcal{A}$ are finite
- One-step dynamics of the environment
  - Probability of each possible pair of next state s' and reward r

$$p(s', r|s, a) = \Pr\{S_{t+1} = s', R_{t+1} = r \mid S_t = s, A_t = a\}$$

  - State-transition probability (T)

$$p(s'|s, a) = \Pr\{S_{t+1} = s' \mid S_t = s, A_t = a\} = \sum_{r \in \mathcal{R}} p(s', r|s, a)$$

  - Expected rewards for state-action pairs

$$r(s, a) = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a] = \sum_{r \in \mathcal{R}} r \sum_{s' \in \mathcal{S}} p(s', r|s, a)$$

  - Expected rewards for state-action-next_state triples

$$r(s, a, s') = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a, S_{t+1} = s'] = \frac{\sum_{r \in \mathcal{R}} r\, p(s', r|s, a)}{p(s'|s, a)}$$

Markov Decision Processes

(Ex) Recycling Robot MDP

– State & action spaces

$$\mathcal{S} = \{\texttt{high}, \texttt{low}\}$$
$$\mathcal{A}(\texttt{high}) = \{\texttt{search}, \texttt{wait}\}$$
$$\mathcal{A}(\texttt{low}) = \{\texttt{search}, \texttt{wait}, \texttt{recharg}$$

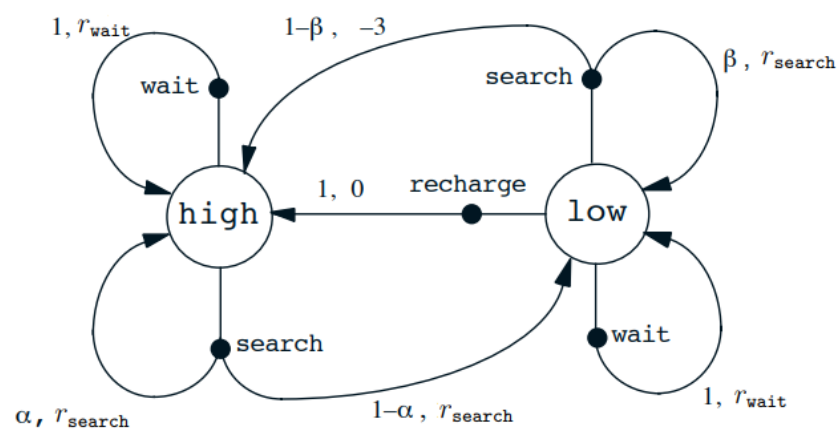
– State transition probability & expected rewards

s	s'	a	$p(s'\|s,a)$	$r(s,a,s')$
high	high	search	α	$r_{\texttt{search}}$
high	low	search	$1-\alpha$	$r_{\texttt{search}}$
low	high	search	$1-\beta$	-3
low	low	search	β	$r_{\texttt{search}}$
high	high	wait	1	$r_{\texttt{wait}}$
high	low	wait	0	$r_{\texttt{wait}}$
low	high	wait	0	$r_{\texttt{wait}}$
low	low	wait	1	$r_{\texttt{wait}}$
low	high	recharge	1	0
low	low	recharge	0	0.

$r_{\texttt{search}}$ = expected no. of cans while searching

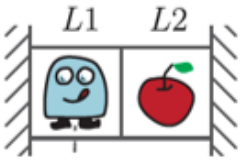
$r_{\texttt{wait}}$ = expected no. of cans while waiting

- Transition graph



Markov Decision Processes

(Ex) 1x2 Grid World



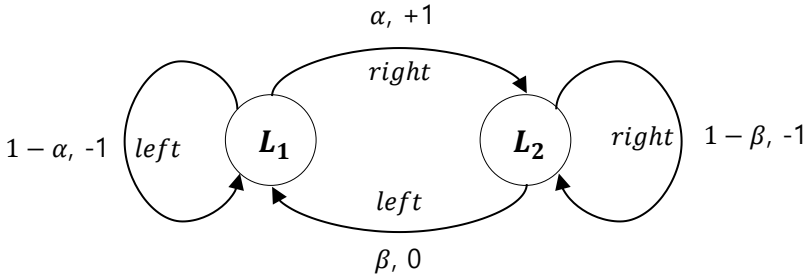
$\mathcal{S} = \{L_1, L_2\}$

$\mathcal{A}(L_1) = \{right, left\}$
$\mathcal{A}(L_2) = \{right, left\}$

- 에이전트는 오른쪽이나 왼쪽으로 이동할 수 있다.

- 에이전트가 $L1$에서 $L2$로 이동하면 사과를 받아 +1의 보상을 얻는다.
- 에이전트가 $L2$에서 $L1$로 이동하면 사과가 다시 생성된다.
- 벽에 부딪히면 −1의 보상을 얻는다. 즉, 벌을 받는다. 예를 들어 에이전트가 $L1$에서 왼쪽으로 이동하면 −1의 보상을, $L2$에서 오른쪽으로 이동해도 마찬가지로 −1의 보상을 얻는다(이때 사과는 다시 생성되지 않는다).
- 지속적 과제, 즉 '끝이 없는' 문제다.

$\boldsymbol{s}$	$\boldsymbol{s'}$	$\boldsymbol{a}$	$\boldsymbol{p(s'\|s,a)}$	$\boldsymbol{r(s,a,s')}$
L_1	L_1	$left$	$1-\alpha$	-1
L_1	L_2	$right$	α	+1
L_2	L_1	$left$	β	0
L_2	L_2	$right$	1-β	-1

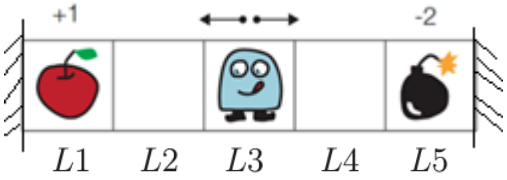


Markov Decision Processes

(Q1) 1x5 Grid World

- State : $\mathcal{S} = \{L_1, \cdots, L_5\}$
- Reward: 사과 (+1), 폭탄 (-2), 빈칸 (0), 벽(0)

⇒ Draw the transition diagram



Value Function

- Value (가치)
  - Measure of estimating *how good* it is
- State-value function: $v_\pi(s)$
  - Value of a state $s \in \mathcal{S}$ under a policy π
    - Policy $\pi(a|s)$ = probability of selecting action $a \in \mathcal{A}$ from state $s \in \mathcal{S}$
  - Expected return when starting in s and thereafter following policy π

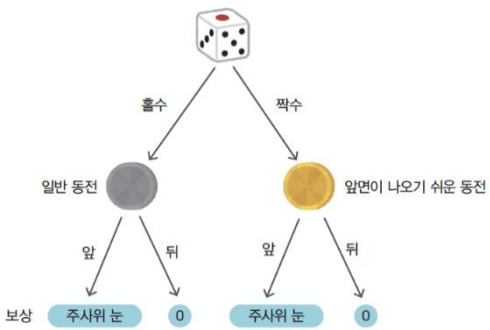
$$\boxed{v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s] = \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \,\middle|\, S_t = s\right]} \qquad v_\pi : \mathcal{S} \to \Re$$

- Action-value function: $q_\pi(s, a)$
  - Value of taking the action $a \in \mathcal{A}$ in state $s \in \mathcal{S}$ under a policy π
  - Expected return when starting in s, taking the action a, and thereafter following policy π

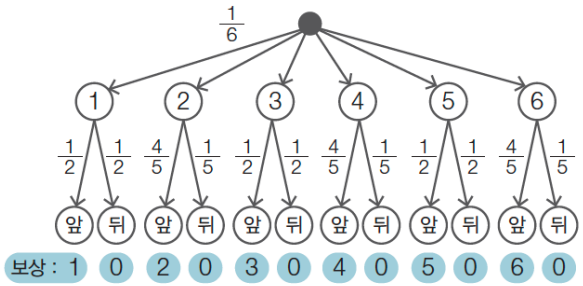
$$\boxed{q_\pi(s, a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a] = \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \,\middle|\, S_t = s, A_t = a\right]} \qquad q_\pi : \mathcal{S} \times \mathcal{A} \to \Re$$

Calculation of Value Function

- Expectation $\mathbb{E}[\cdot]$
  (ex) 주사위 던지기 → 동전 던지기



backup diagram



  - Variables: 주사위 $x \in \{1, \dots, 6\}$, 동전 $y \in \{0(\text{뒤}), 1(\text{앞})\}$
  - Probability: $p(x, y) = p(x)\, p(y|x)$ (ex) $p(x = 4, y = \text{앞}) = \frac{1}{6} \cdot \frac{4}{5} = \frac{4}{30}$
  - Reward: $r(x, y)$ (ex) $r(x = 4, y = \text{앞}) = 4$
  - Expectation

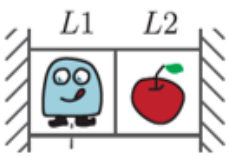
$$\mathbb{E}[r(x,y)] = \sum_x \sum_y p(x,y)\, r(x,y) = \sum_x \sum_y p(x)p(y|x)r(x,y)$$

$$\left(\frac{1}{6}\cdot\frac{1}{2}\cdot 1\right)+\left(\frac{1}{6}\cdot\frac{1}{2}\cdot 0\right)+\left(\frac{1}{6}\cdot\frac{4}{5}\cdot 2\right)+\left(\frac{1}{6}\cdot\frac{1}{5}\cdot 0\right)+\left(\frac{1}{6}\cdot\frac{1}{2}\cdot 3\right)+\left(\frac{1}{6}\cdot\frac{1}{2}\cdot 0\right)+$$
$$\left(\frac{1}{6}\cdot\frac{4}{5}\cdot 4\right)+\left(\frac{1}{6}\cdot\frac{1}{5}\cdot 0\right)+\left(\frac{1}{6}\cdot\frac{1}{2}\cdot 5\right)+\left(\frac{1}{6}\cdot\frac{1}{2}\cdot 0\right)+\left(\frac{1}{6}\cdot\frac{4}{5}\cdot 6\right)+\left(\frac{1}{6}\cdot\frac{1}{5}\cdot 0\right)$$
$$= 2.35$$

Calculation of Value Function

$$v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s] = \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \;\middle|\; S_t = s\right]$$

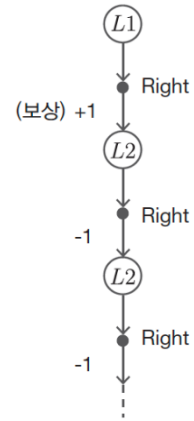
(Ex) 1x2 Grid World



1) Policy $\pi 1(a|s) = \mu 1(s)$ (deterministic policy)

a \ s	L_1	L_2
$left$	0	0
$right$	1	1

$\gamma = 0.9$



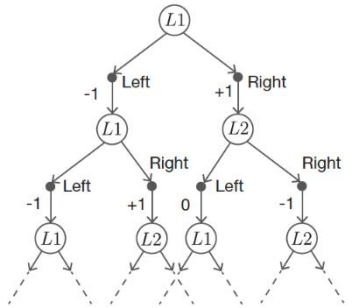
$$\begin{aligned} v_{\mu_1}(S = L1) &= 1 + 0.9 \cdot (-1) + 0.9^2 \cdot (-1) + \cdots \\ &= 1 - 0.9(1 + 0.9 + 0.9^2 + \cdots) \\ &= 1 - \frac{0.9}{1 - 0.9} \\ &= -8 \end{aligned}$$

$$\begin{aligned} v_{\mu_1}(s = L2) &= -1 + 0.9 \cdot (-1) + 0.9^2 \cdot (-1) + \cdots \\ &= -1 - 0.9(1 + 0.9 + 0.9^2 + \cdots) \\ &= -1 - \frac{0.9}{1 - 0.9} \\ &= -10 \end{aligned}$$

2) Policy $\pi 1(\mathrm{a}|\mathrm{s})$ (stochastic policy)

a \ s	L_1	L_2
$left$	0.5	0.5
$right$	0.5	0.5

$\gamma = 0.9$



$$\begin{aligned} v_{\pi 1}(L1) &= 0.5((-1) + 0.9 \times (0.5((-1) + \cdots)) \\ &\quad + 0.5(1 + 0.9 \times (0.5(0 + \cdots))) \\ &= ????? \end{aligned}$$

➢ Value function 의 계산: (deterministic policy) 계산 용이
(stochastic policy) 계산 힘듬

Quiz

(Q2) For 1x2 grid world, derive the $v_{\mu 2}(s), v_{\mu 3}(s)$ and $v_{\mu 4}(s)$ $(\gamma = 0.9)$

$\mu 1$

	L_1	L_2
$left$	0	0
$right$	1	1

$\mu 2$

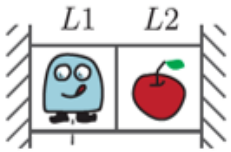
	L_1	L_2
$left$	0	1
$right$	1	0

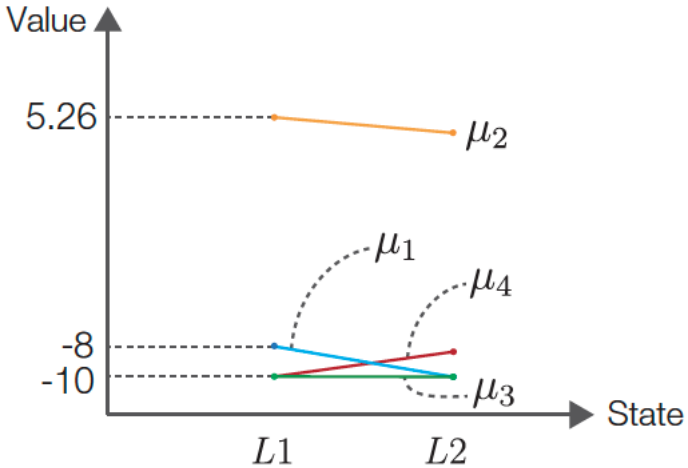
$\mu 3$

	L_1	L_2
$left$	1	0
$right$	0	1

$\mu 4$

	L_1	L_2
$left$	1	1
$right$	0	0





Bellman Equation – State Value

- Return (수익)

$$G_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \cdots$$

$$G_{t+1} = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots$$

R_t: Reward (보상)

$$\Rightarrow \quad G_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \cdots$$

$$= R_t + \gamma (R_{t+1} + \gamma R_{t+2} + \cdots)$$

$$= \underline{R_t + \gamma G_{t+1}}$$

- State Value (상태 가치) – stochastic case

$v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$ policy π , state=s 의 Expected Return (기대수익)

$$v_\pi(s) = \mathbb{E}_\pi[G_t | S_t = s]$$

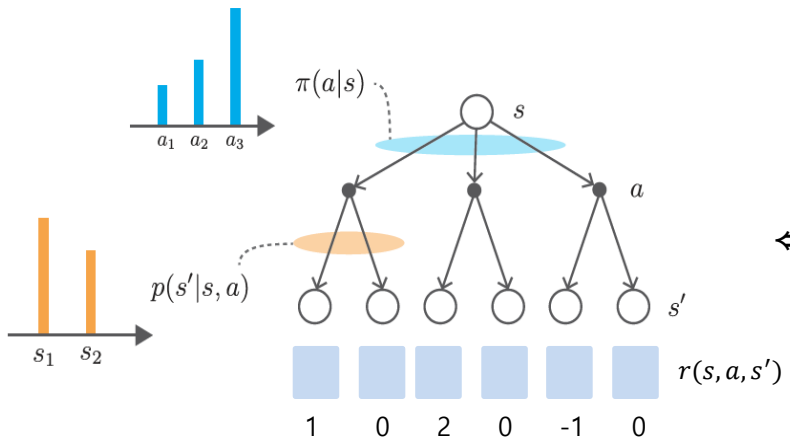
$$= \mathbb{E}_\pi[R_t + \gamma G_{t+1} | S_t = s]$$

$$= \underbrace{\mathbb{E}_\pi[R_t | S_t = s]}_{①} + \gamma \underbrace{\mathbb{E}_\pi[G_{t+1} | S_t = s]}_{②}$$

$\leftarrow \mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$

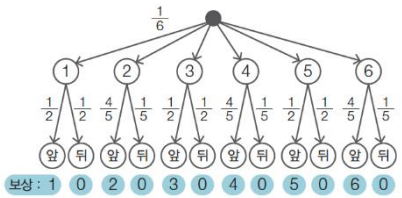
Bellman Equation – State Value

① $\mathbb{E}_\pi[R_t \mid S_t = s]$



⇔

(ex) 주사위-동전 던지기



– Policy $\pi(a \mid s)$

$\pi(a = a_1 \mid s) = 0.2$

$\pi(a = a_2 \mid s) = 0.3$

$\pi(a = a_3 \mid s) = 0.5$

- State transition probability $p(s' \mid s, a)$

$p(s' = s_1 \mid s, a = a_1) = 0.6$

$p(s' = s_2 \mid s, a = a_1) = 0.4$

– State-value $\mathbb{E}_\pi[R_t \mid S_t = s]$

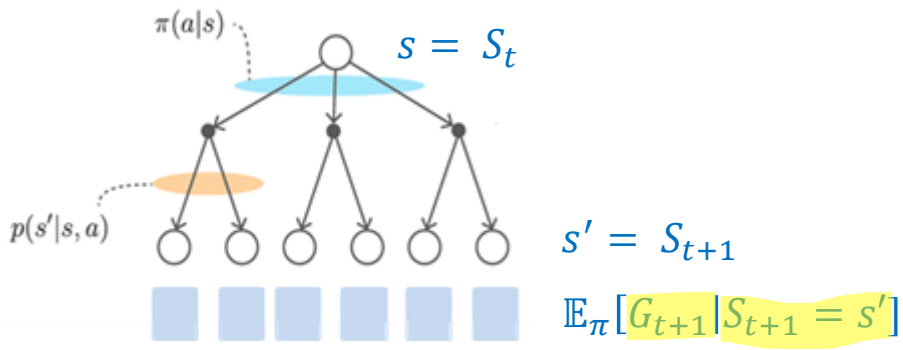
$$\begin{aligned}\mathbb{E}_\pi[R_t \mid S_t = s] = {} & \pi(a_1|s)\, p(s_1|s,a_1) r(s,a_1,s_1) + \pi(a_1|s)\, p(s_2|s,a_1) r(s,a_1,s_2) \\ & + \pi(a_2|s)\, p(s_1|s,a_2) r(s,a_2,s_1) + \pi(a_2|s)\, p(s_2|s,a_2) r(s,a_2,s_2) \\ & + \pi(a_3|s)\, p(s_1|s,a_3) r(s,a_3,s_1) + \pi(a_3|s)\, p(s_3|s,a_2) r(s,a_3,s_2)\end{aligned}$$

$$\begin{aligned} = {} & 0.2 \times 0.6 \times 1 + 0.2 \times 0.4 \times 0 \\ & + 0.3 \times 0.6 \times 2 + 0.3 \times 0.4 \times 0 \\ & + 0.5 \times 0.6 \times (-1) + 0.5 \times 0.4 \times 0 \\ = {} & 0.18 \end{aligned}$$

$$\mathbb{E}_\pi[R_t \mid S_t = s] = \sum_a \sum_{s'} \pi(a|s)\, p(s'|s,a)\, r(s,a,s') = \sum_{a,s'} \pi(a|s)\, p(s'|s,a)\, r(s,a,s')$$

Bellman Equation – State Value

② $\mathbb{E}_\pi[G_{t+1} \mid S_t = s]$



$$\mathbb{E}_\pi[G_{t+1} \mid S_t = s] = \sum_{a,s'} \pi(a|s)\, p(s'|s,a)\mathbb{E}_\pi[G_{t+1} \mid S_{t+1} = s']$$

Bellman Equation – State Value

- State-Value $v_\pi(s)$

$$v_\pi(s) \equiv \mathbb{E}_\pi[G_t|S_t = s]$$

$$= \mathbb{E}_\pi[R_t|S_t = s] + \gamma\mathbb{E}[G_{t+1}|S_{t+1} = s]$$

$$= \underbrace{\sum_{a,s'} \pi(a|s)\, p(s'|s,a)\, r(s,a,s')}_{①} + \gamma \underbrace{\sum_{a,s'} \pi(a|s)\, p(s'|s,a)\mathbb{E}_\pi[G_{t+1} \mid S_{t+1} = s']}_{②}$$

$$= \sum_{a,s'} \pi(a|s)\, p(s'|s,a)\, \{r(s,a,s') + \gamma \underbrace{\mathbb{E}_\pi[G_{t+1} \mid S_{t+1} = s']}_{v_\pi(s')}\}$$

- Bellman Equation

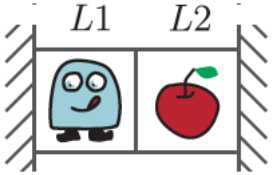
$$\boxed{v_\pi(s) = \sum_{a,s'} \pi(a|s)\, p(s'|s,a)\, \{r(s,a,s') + \gamma v_\pi(s')\}}$$

  – Recursive relationships between state-value at s and next state s'

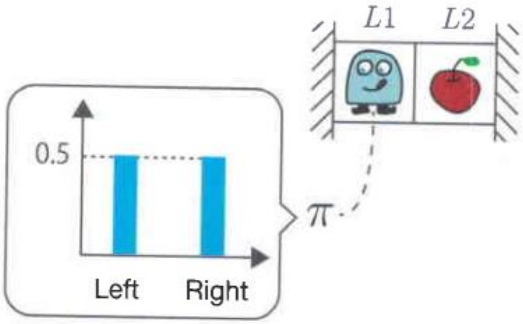
  – $v_\pi(s),\ s = s_1, \cdots . s_n$ 에 대한 1차 연립방정식 문제 (deterministic & stochastic case)

Example

- 2-Grid World



- 에이전트는 오른쪽이나 왼쪽으로 이동할 수 있다. *Stochastic policy*



- 상태 전이는 결정적이다.
- 에이전트가 $L1$에서 $L2$로 이동하면 사과를 받아 +1의 보상을 얻는다.
- 에이전트가 $L2$에서 $L1$로 이동하면 사과가 다시 생성된다.
- 벽에 부딪히면 −1의 보상을 얻는다. 즉, 벌을 받는다. 예를 들어 에이전트가 $L1$에서 왼쪽으로 이동하면 −1의 보상을, $L2$에서 오른쪽으로 이동해도 마찬가지로 −1의 보상을 얻는다(이때 사과는 다시 생성되지 않는다).
- 지속적 과제, 즉 '끝이 없는' 문제다. *Continuous task*

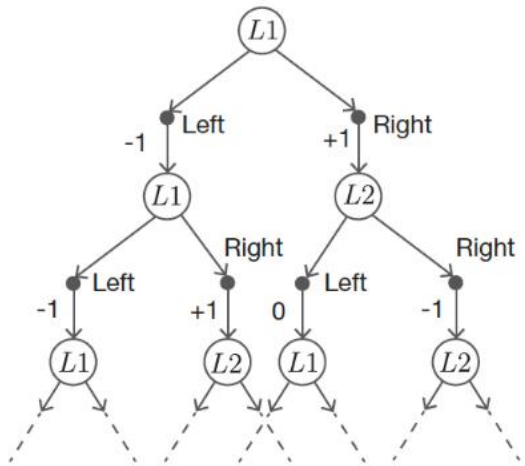
Example

- Non-Bellman Equation
  - Infinite-series computation problem
  - Hard to solve

$\pi(a|s)$

a \ s	L_1	L_2
$left$	0.5	0.5
$right$	0.5	0.5

$\gamma = 0.9$



$$v_{\pi 1}(L1) = 0.5((-1) + 0.9 \times (0.5((-1) + \cdots))$$

$$+ 0.5(1 + 0.9 \times (0.5(0 + \cdots)))$$

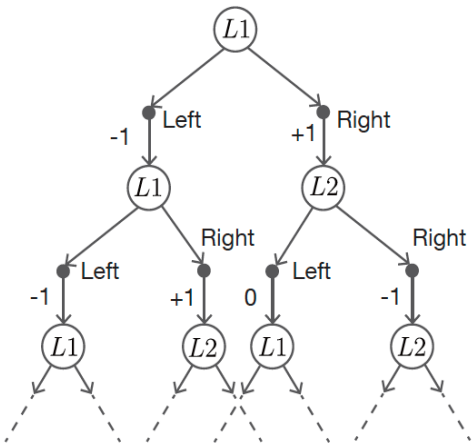
$$= ?????$$

Example

- Bellman equation
  - Systems of linear equations (1차 연립방정식)

$$v_\pi(s) = \sum_{a,s'} \pi(a \mid s)\, p(s' \mid s,a)\{r(s,a,s') + \gamma v_\pi(s')\}$$

$$= \sum_{a} \pi(a \mid s) \sum_{s'} p(s' \mid s,a)\{r(s,a,s') + \gamma v_\pi(s')\}$$



(backup diagram)

- Policy $\pi(a|s)$

$\pi(LEFT \mid s = L1,) = \pi(LEFT \mid s = L2) = 0.5$

$\pi(RIGHT \mid s = L1,) = \pi(RIGHT \mid s = L2) = 0.5$

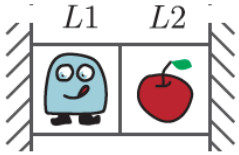
- State transition probability $p(s'|s,a)$: Deterministic transition 인 경우

$s' = f(s,a)$이면 $p(s'|s,a) = 1$

$s' \neq f(s,a)$이면 $p(s'|s,a) = 0$

(ex) $p(s' = L2 \mid s = L1, a = RIGHT)$=1

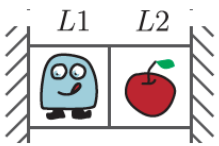
$p(s' = L1 \mid s = L2, a = RIGHT)$=0



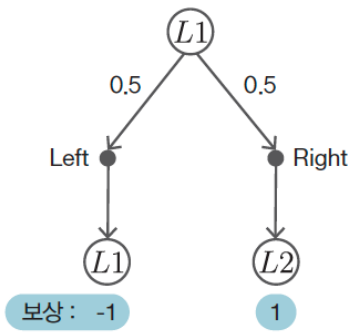
$$\boxed{v_\pi(s) = \sum_{a} \pi(a \mid s)\{r(s,a,s') + \gamma v_\pi(s')\}}$$ when $s' = f(s,a)$

Example

- Bellman equation

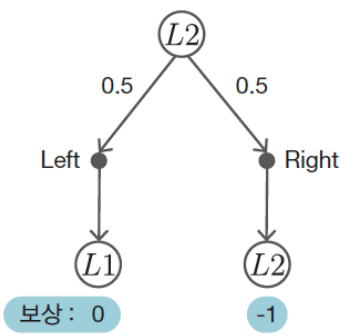


$$v_\pi(s) = \sum_a \pi(a \mid s)\{r(s,a,s') + \gamma v_\pi(s')\} \qquad \gamma = 0.9$$



$$v_\pi(L1) = \overbrace{0.5\{-1+0.9v_\pi(L1)\}}^{a\,=\,LEFT} + \overbrace{0.5\{1+0.9v_\pi(L2)\}}^{a\,=\,RIGHT}$$

$$-0.55v_\pi(L1) + 0.45v_\pi(L2) = 0 \quad \text{①}$$



$$v_\pi(L2) = 0.5\{0+0.9v_\pi(L1)\} + 0.5\{-1+0.9v_\pi(L2)\}$$

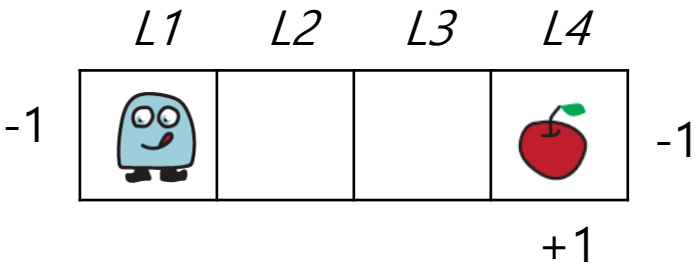
$$0.45v_\pi(L1) - 0.55v_\pi(L2) = 0.5 \quad \text{②}$$

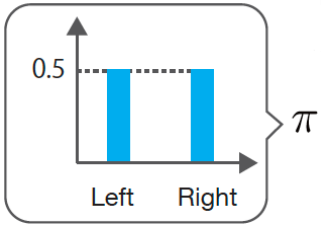
①, ② ⇒ 연립방정식

$$\boxed{\begin{cases} v_\pi(L1) = -2.25 \\ v_\pi(L2) = -2.75 \end{cases}}$$

Quiz

(Q3) 앞 Example 의 2-Grid World 문제를 4-Grid World 문제로 확장하는 경우, state value 에 대한 Bellman equation 을 적고, 각 state value 를 구하라.





$\gamma = 0.9$

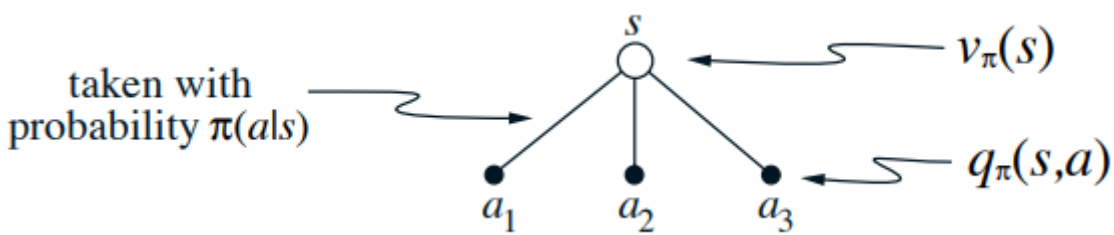
Bellman Equation – Action Value

- Action Value (행동 가치)

$$q_\pi(s,a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a\,]$$

policy=π , state=s, action=a 일 때 기대 수익
Q Function

(cf) $v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s,]$: state value



$$v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s] = \sum_a \pi(a \mid s)\, \mathbb{E}[G_t | S_t = s, A_t = a]$$

$$= \sum_a \pi(a \mid s)\, q_\pi(s,a) \qquad (\bigstar)$$

Bellman Equation – Action Value

- Bellman Equation

$$
\begin{aligned}
q_\pi(s,a) &= \mathbb{E}_\pi[G_t \mid S_t = s,\ A_t = a] \\
&= \mathbb{E}_\pi[R_t + \gamma G_{t+1} \mid S_t = s,\ A_t = a] \\
&= \mathbb{E}_\pi[R_t \mid S_t = s,\ A_t = a] + \gamma \mathbb{E}_\pi[G_{t+1} \mid S_t = s,\ A_t = a] \\
&= \sum_{s'} p(s' \mid s,a) r(s,a,s') + \gamma \sum_{s'} p(s' \mid s,a) \mathbb{E}_\pi[G_{t+1} \mid S_{t+1} = s'] \\
&= \sum_{s'} p(s' \mid s,a) \{ r(s,a,s') + \gamma \mathbb{E}_\pi[G_{t+1} \mid S_{t+1} = s'] \} \\
&= \underbrace{\sum_{s'} p(s' \mid s,a) \{ r(s,a,s') + \gamma v_\pi(s') \}}_{(\bigstar\bigstar)} \longleftarrow v_\pi(s) = \sum_a \pi(a \mid s) q_\pi(s,a) \ (\bigstar)
\end{aligned}
$$

$$
\boxed{q_\pi(s,a) = \sum_{s'} p(s' \mid s,a) \left\{ r(s,a,s') + \gamma \sum_{a'} \pi(a' \mid s') q_\pi(s',a') \right\}}
$$

☞ $S_t = s,\ A_t = a$ 에 대한 기대수익 $q_\pi(s,a)$

$S_{t+1} = s',\ A_{t+1} = a'$ 에 대한 기대수익 $q_\pi(s',a')$

방정식

Optimal Value Function & Policy

- Optimal state-value function
  - Maximum state-value function overall polices

$$\boxed{v_*(s) = \max_{\pi} v_\pi(s)}$$

- Optimal action-value function
  - Maximum action-value function overall polices

$$\boxed{q_*(s,a) = \max_{\pi} q_\pi(s,a)}$$

- Optimal action μ_*

$$\boxed{\mu_*(s) = \arg\max_{a} q_*(s,a)}$$

- Optimal policy π_*

$$\boxed{\pi_*(a|s) = \begin{cases} 1, & \text{if } a = \arg\max_{a \in \mathcal{A}} q_*(s,a) \\ 0, & \text{otherwise.} \end{cases}}$$

Bellman Optimality Equation

- Optimal State-Value
  - Bellman equation

$$v_\pi(s) = \sum_{a,s'} \pi(a \mid s) p(s' \mid s,a) \{r(s,a,s') + \gamma v_\pi(s')\}$$

$$= \sum_{a} \pi(a \mid s) \sum_{s'} p(s' \mid s,a) \{r(s,a,s') + \gamma v_\pi(s')\}$$

  - Optimal policy $\pi_*(a|s)$ & optimal state-value $v_*(s)$
    - State value $v_\pi(s)$ 를 최대화하는 policy $\pi(a|s)$
  - Bellman optimality equation

$$v_*(s) = \sum_{a} \pi_*(a \mid s) \sum_{s'} p(s' \mid s,a) \{r(s,a,s') + \gamma v_*(s')\}$$

$$\Rightarrow \boxed{v_*(s) = \max_a \sum_{s'} p(s' \mid s,a) \{r(s,a,s') + \gamma v_*(s')\}}$$

최적화 문제 !

(proof) $v_*(s) = \sum_a \pi_*(a|s) \sum_{s'} p(s'|s,a) \{r(s,a,s') + \gamma v_*(s')\}$

$$= \begin{cases} -2 & (a_1) \\ 0 & (a_2) \\ 4 & (a_3) \end{cases}$$

$\rightarrow$ $\pi(a_1|s) = 0,$ $\pi(a_2|s) = 0,$ $\pi(a_3|s) = 1,$ $\rightarrow$ $v_*(s) = \max\{-2,0,4\} = 4$

$\downarrow$

$$v_*(s) = \max_a \sum_{s'} p(s' \mid s,a) \{r(s,a,s') + \gamma v_*(s')\}$$

Bellman Optimality Equation

- Optimal Action-Value
  - Bellman equation

$$q_\pi(s,a) = \sum_{s'} p(s' \mid s,a)\left\{r(s,a,s') + \gamma \sum_{a'} \pi(a' \mid s') q_\pi(s',a')\right\}$$

  - Optimal policy: : $\pi_*(a|s)$, optimal action-state: $q_*(s,a)$
  - Bellman optimality equation

$$q_*(s,a) = \sum_{s'} p(s' \mid s,a)\left\{r(s,a,s') + \gamma \sum_{a'} \pi_*(a' \mid s') q_*(s',a')\right\}$$

$$\Rightarrow \quad \boxed{q_*(s,a) = \sum_{s'} p(s' \mid s,a)\left\{r(s,a,s') + \gamma \max_{a'} q_*(s',a')\right\}}$$

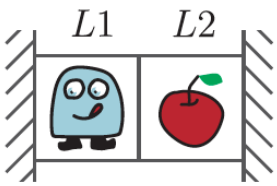
최적화 문제 !

- Optimal Policy
  - optimal action $\mu_*(s) = \arg\max_a q_*(s,a)$

$$\Rightarrow \quad \boxed{\pi_*(a|s) = \begin{cases} 1 & if\ a = \mu_*(s) \\ 0 & otherwise \end{cases}}$$

Example 1

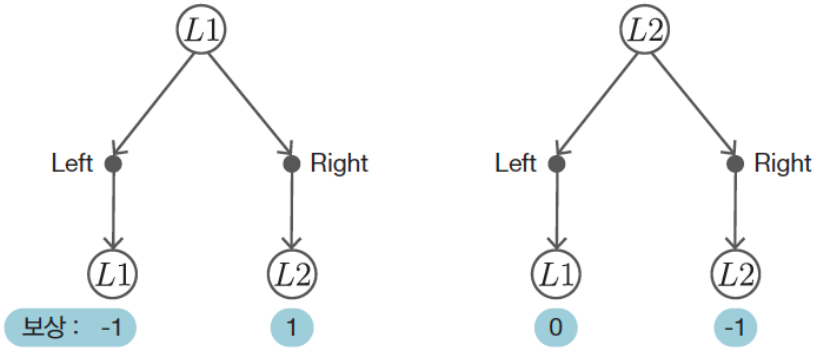
- 2-Grid World



  – Bellman optimality equation: optimal state value

$$v_*(s) = \max_a \sum_{s'} p(s' \mid s,a)\{r(s,a,s') + \gamma v_*(s')\}$$

$s' = f(s,a)$일 때 $\quad v_*(s) = \max_a \{r(s,a,s') + \gamma v_*(s')\}$ $\qquad \gamma = 0.9$



$$v_*(L1) = \max\begin{Bmatrix} -1+0.9v_*(L1), \\ 1+0.9v_*(L2) \end{Bmatrix}$$

$$v_*(L2) = \max\begin{Bmatrix} 0.9v_*(L1), \\ -1+0.9v_*(L2) \end{Bmatrix}$$

$\Rightarrow$ $v_*(L1) = 5.26$

$v_*(L2) = 4.73$

$v_*(s)$

5.26	4.73
$L1$	$L2$

Example 1

- Optimal policy
  - Action value from state value

$$q_\pi(s,a) = \sum_{s'} p(s'|s,a)\,\{r(s,a,s') + r v_\pi(s')\} \qquad (\bigstar\bigstar)$$

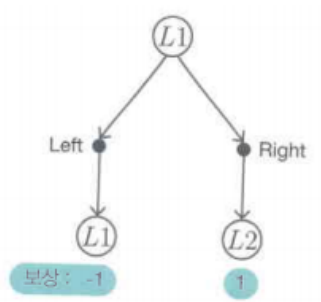
  - Optimal action

$$\mu_*(s) = \arg\max_a q_\pi(s,a)$$

$$= \arg\max_a \sum_{s'} p(s'|s,a)\,\{r(s,a,s') + \gamma v_*(s')\}$$

$v_*(s)$

5.26	4.73
$L1$	$L2$

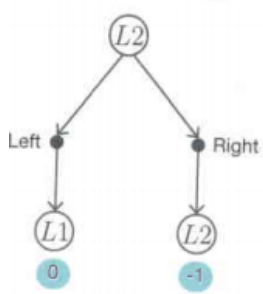


$$\sum_{s'} p(s'|s,a)\,\{r(s,a,s') + \gamma v_*(s')\}$$

$$= \begin{cases} -1 + 0.9\, v_*(L1), & a = LEFT \\ 1 + 0.9\, v_*(L2), & a = RIGHT \end{cases}$$

$$= \begin{cases} 3.734, & a = LEFT \\ 5.257, & a = RIGHT \end{cases}$$

$$\Rightarrow \quad \mu_*(L1) = RIGHT$$



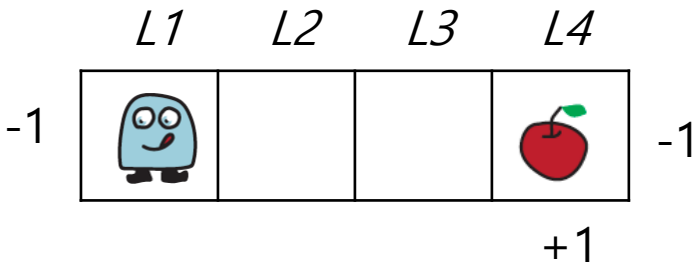
$$= \begin{cases} 0.9\, v_*(L1), & a = LEFT \\ -1 + 0.9\, v_*(L2), & a = RIGHT \end{cases}$$

$$= \begin{cases} 4.734, & a = LEFT \\ 3.257, & a = RIGHT \end{cases}$$

$$\Rightarrow \quad \mu_*(L2) = LEFT$$

Quiz

(Q4) 앞 Example 의 2-grid World 문제를 확장한 4-Grid World 문제에 대하여, Bellman optimality equation 을 유도하라.



Example 2

- Recycling Robot
  - Markov decision process
    - State & action spaces

$$\mathcal{S} = \{\texttt{high}, \texttt{low}\}$$
$$\mathcal{A}(\texttt{high}) = \{\texttt{search}, \texttt{wait}\}$$
$$\mathcal{A}(\texttt{low}) = \{\texttt{search}, \texttt{wait}, \texttt{recharg}$$

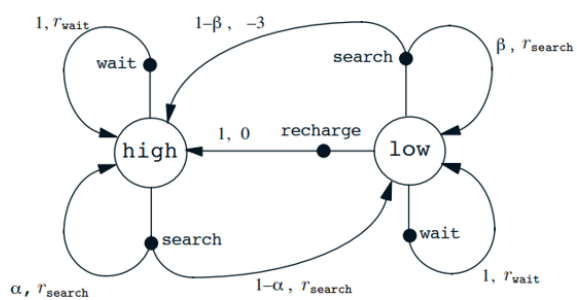
    - State transition probability & expected rewards

s	s'	a	$p(s'\|s,a)$	$r(s,a,s')$
high	high	search	α	$r_{\texttt{search}}$
high	low	search	$1-\alpha$	$r_{\texttt{search}}$
low	high	search	$1-\beta$	-3
low	low	search	β	$r_{\texttt{search}}$
high	high	wait	1	$r_{\texttt{wait}}$
high	low	wait	0	$r_{\texttt{wait}}$
low	high	wait	0	$r_{\texttt{wait}}$
low	low	wait	1	$r_{\texttt{wait}}$
low	high	recharge	1	0
low	low	recharge	0	0.

$r_{\texttt{search}}$ = expected no. of cans while searching

$r_{\texttt{wait}}$ = expected no. of cans while waiting

- Transition graph



  - Bellman optimality equation: optimal state value

$$\begin{aligned} v_*(\texttt{h}) &= \max\left\{ \begin{array}{l} p(\texttt{h}|\texttt{h},\texttt{s})[r(\texttt{h},\texttt{s},\texttt{h}) + \gamma v_*(\texttt{h})] + p(\texttt{l}|\texttt{h},\texttt{s})[r(\texttt{h},\texttt{s},\texttt{l}) + \gamma v_*(\texttt{l})], \\ p(\texttt{h}|\texttt{h},\texttt{w})[r(\texttt{h},\texttt{w},\texttt{h}) + \gamma v_*(\texttt{h})] + p(\texttt{l}|\texttt{h},\texttt{w})[r(\texttt{h},\texttt{w},\texttt{l}) + \gamma v_*(\texttt{l})] \end{array} \right\} \\ &= \max\left\{ \begin{array}{l} \alpha[r_\texttt{s} + \gamma v_*(\texttt{h})] + (1-\alpha)[r_\texttt{s} + \gamma v_*(\texttt{l})], \\ 1[r_\texttt{w} + \gamma v_*(\texttt{h})] + 0[r_\texttt{w} + \gamma v_*(\texttt{l})] \end{array} \right\} \\ &= \max\left\{ \begin{array}{l} r_\texttt{s} + \gamma[\alpha v_*(\texttt{h}) + (1-\alpha) v_*(\texttt{l})], \\ r_\texttt{w} + \gamma v_*(\texttt{h}) \end{array} \right\}. \end{aligned}$$

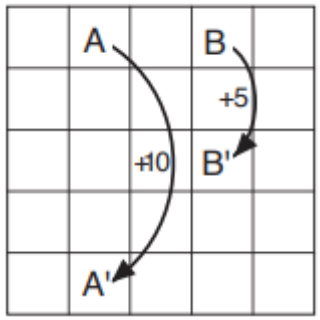
$$v_*(\texttt{l}) = \max\left\{ \begin{array}{l} \beta r_\texttt{s} - 3(1-\beta) + \gamma[(1-\beta) v_*(\texttt{h}) + \beta v_*(\texttt{l})] \\ r_\texttt{w} + \gamma v_*(\texttt{l}), \\ \gamma v_*(\texttt{h}) \end{array} \right\}$$

최적화 문제 !! How to Solve ?? ☞ "Dynamic Programming"

Example 3

- 5x5 Grid World

Example 3.8: Gridworld Figure 3.5a uses a rectangular grid to illustrate value functions for a simple finite MDP. The cells of the grid correspond to the states of the environment. At each cell, four actions are possible: north, south, east, and west, which deterministically cause the agent to move one cell in the respective direction on the grid. Actions that would take the agent off the grid leave its location unchanged, but also result in a reward of -1. Other actions result in a reward of 0, except those that move the agent out of the special states A and B. From state A, all four actions yield a reward of $+10$ and take the agent to A′. From state B, all actions yield a reward of $+5$ and take the agent to B′.



Actions

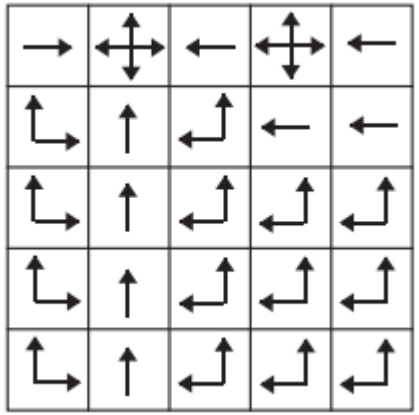
$v_\pi(s)$ $(\gamma = 0.9)$

3.3	8.8	4.4	5.3	1.5
1.5	3.0	2.3	1.9	0.5
0.1	0.7	0.7	0.4	-0.4
-1.0	-0.4	-0.4	-0.6	-1.2
-1.9	-1.3	-1.2	-1.4	-2.0

$v_*(s)$

22.0	24.4	22.0	19.4	17.5
19.8	22.0	19.8	17.8	16.0
17.8	19.8	17.8	16.0	14.4
16.0	17.8	16.0	14.4	13.0
14.4	16.0	14.4	13.0	11.7

$\pi_*(a|s)$



요약

- Value Function ☞ Bellman Equation

$$v_\pi(s) = \sum_{a,s'} \pi(a \mid s)\, p(s' \mid s,a)\{r(s,a,s') + \gamma v_\pi(s')\}$$

$$q_\pi(s,a) = \sum_{s'} p(s' \mid s,a)\left\{r(s,a,s') + \gamma \sum_{a'} \pi(a' \mid s')\, q_\pi(s',a')\right\}$$

연립방정식 문제

- Optimal Value Function ☞ Bellman Optimality Equation

$$v_*(s) = \max_a \sum_{s'} p(s' \mid s,a)\{r(s,a,s') + \gamma v_*(s')\}$$

$$q_*(s,a) = \sum_{s'} p(s' \mid s,a)\left\{r(s,a,s') + \gamma \max_{a'} q_*(s',a')\right\}$$

$$\mu_*(s) = \operatorname*{argmax}_a q_*(s,a)$$

최적화 문제
(Dynamic Programming)